

Software Engineering Project Proposal

Smart House Design & Cost Recommendation System

Team Name: FivePlusOne Devs

CSC 312 — Software Engineering Group Project

1. Project Abstract & Problem Statement

Problem Definition

Prospective homeowners often struggle to match house designs with their plot size and budget, which leads to confusion and unreliable cost estimates during the early planning stages of residential construction.

Core Aim

This project proposes a recommendation system that suggests suitable house designs based on a user's plot size and construction budget, and displays the associated estimated construction costs. By streamlining design selection and cost estimation, the system helps homeowners make more informed and confident decisions.

2. Stakeholder Identification

Primary Stakeholders

- **Homeowners / Clients:** End-users who input their plot size and budget to receive matching house design recommendations and cost estimates, enabling more confident planning decisions.
- **System Administrators:** Manage user accounts, house plan data, and system content to keep recommendations accurate and up to date.

Secondary Stakeholders

- **Architects & House Designers:** Supply house designs and technical specifications that populate the system's recommendation database.
- **Contractors:** Provide cost estimation data used to calculate and refine the system's construction cost outputs.

3. Software Process Model & Software Engineering Practices

Process Model

We will use the Agile Scrum process model, developing the system through iterative sprints with regular feedback cycles. Scrum is well suited to this project because requirements (such as design and cost-estimation logic) are likely to evolve as we receive stakeholder feedback, and the fixed academic timeline (Term 3 and Term 4) maps naturally onto a series of short, time-boxed sprints with incremental, demonstrable deliverables.

Process Elements

SE Principles

- **User-Centered Design:** Requirements and interface decisions are driven by the needs of homeowners, ensuring recommendations and workflows remain intuitive and relevant.

- **Modular Architecture:** The system is built from independent, loosely coupled modules (e.g., recommendation engine, cost estimator, admin management) to simplify testing, maintenance, and future scaling.

SE Methods / Techniques

- **UML Use Case & Class Diagrams:** Used to model system actors, interactions, and object structures before implementation begins.
- **Requirement Analysis:** Structured elicitation and documentation of functional and non-functional requirements to guide sprint planning.
- **Low-Fidelity Prototyping:** Early wireframes and mockups used to validate UI/UX concepts with stakeholders before development.

SE Goals (ISO 25010)

- **Usability:** The system will provide a clear, responsive interface that allows users to easily input data and interpret recommendations.
- **Reliability:** The system will consistently deliver accurate recommendations and remain available, with fault tolerance built into core services.

SE Umbrella Activities

- **Quality Assurance & Testing:** Continuous testing (unit, integration, and user acceptance) throughout each sprint to catch defects early.
- **Risk Management:** Ongoing identification and mitigation of risks such as scope creep and technical integration issues.
- **Version Control (Git):** All code changes are tracked, reviewed, and merged through Git to maintain a stable, auditable codebase.

Toolchain

- **Version Control:** Git and GitHub for source control, branching, and collaboration.
- **Project Tracking:** Jira or Trello for sprint planning, backlog management, and task tracking.
- **UI/UX Design:** Figma for wireframing and prototyping the user interface.
- **Development Environment:** Visual Studio Code as the primary IDE.
- **Database:** MySQL or PostgreSQL for persistent storage of user, design, and cost data.

4. Project Scope & Requirements

The scope below has been kept realistic for delivery across Term 3 and Term 4 of the course.

Functional Requirements

- User registration and login.
- Input of plot size and construction budget.
- Browsing and searching available house designs.
- Viewing recommended house plans that match user inputs.
- Display of estimated construction costs for selected designs.
- Admin functionality to manage user accounts and house plan data.

Non-Functional Requirements

- Secure authentication for all user accounts.
- Fast recommendation response times.
- Responsive UI that works across desktop and mobile devices.
- Reliable system availability.
- Maintainable and scalable architecture to support future growth.